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Objectives: At the end of the course, the students will learn about reinforcement learning and its applications in different areas of engineering.

Course contents:

1. **Formulation of Reinforcement learning problem:** Rewards and returns, Markov Decision Processes, Value functions, Optimality and approximation, value iteration, policy iteration, asynchronous DP, generalized policy iteration, policy evaluation, roll outs, on-policy and off-policy learning, importance sampling
2. **Temporal Difference learning:** TD prediction, SARSA, Q-learning, R-learning, n-step TD prediction, TD (λ), Q (λ), SARSA (λ)
3. **Function Approximation:** Value prediction, Linear function approximation, NN-based function approximation, instability issues
4. **Policy Gradient methods:** REINFORCE algorithm, exact gradient methods, estimating gradients, approximate policy gradient algorithms, actor-critic methods.
5. **Multi-Armed Bandits:** Explore vs Exploit dilemma, Stochastic setting - epsilon greedy, successive elimination, UCB, Thompson Sampling Adversarial setting - EXP3, Contextual bandits - EXP4, linUCB
6. **Online Learning:** Learning from expert advice, definition of regret, Halving, Hedge, Perceptron, Winnow, Follow the leader, Follow the regularized leader, Follow the perturbed leader, Online Gradient Descent

Textbooks:

1. D.P.Bertsekas and J.N.Tsitsiklis, Neuro-Dynamic Programming, Athena Scientific, 1996
 2. R.S.Sutton and A.G.Barto, Reinforcement Learning: An Introduction, MIT Press, 2018
 3. Lecture notes
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